# Reliable USB File Transfer System


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## Abstract


This project presents a reliable USB file transfer system that prevents data loss during unexpected disconnections. Unlike traditional USB transfers that must restart after a failure, this system automatically detects interruptions, saves progress, and resumes the transfer from the last completed point once the connection is restored. This approach saves time, improves efficiency, and enhances user experience, making it especially suitable for large file transfers and unstable connection environments.


## Introduction

- USB file transfer is widely used to move data between computers and external storage devices.
- Traditional USB transfers are not fault-tolerant; any interruption requires restarting the entire process.
- This issue is especially problematic for large file transfers.
- Our project introduces a reliable USB file transfer system with auto-resume support.
- The system tracks transfer progress and resumes from the last successful point after reconnection.
- This approach saves time, improves efficiency, and provides a more dependable user experience.

## Problem Domain

1. Frequent USB disconnections cause file transfer failures, wasting time and effort.
2. Traditional systems cannot resume progress, forcing users to restart large transfers.
3. There is a need for a fault-tolerant method that ensures reliable, uninterrupted data transfer.

## Motivation

1. Users face frustration when large file transfers restart after minor interruptions.
2. A reliable resume mechanism can save significant time and improve efficiency.
3. Providing a smoother, fault-tolerant transfer experience enhances overall user satisfaction.

## Objective

1. To develop a reliable USB file transfer system that prevents full restart after interruptions.
2. To implement an automatic resume feature that continues transfers from the last successful point.
3. To enhance user experience by ensuring faster, efficient, and fault-tolerant file transfer.

## Literature Review

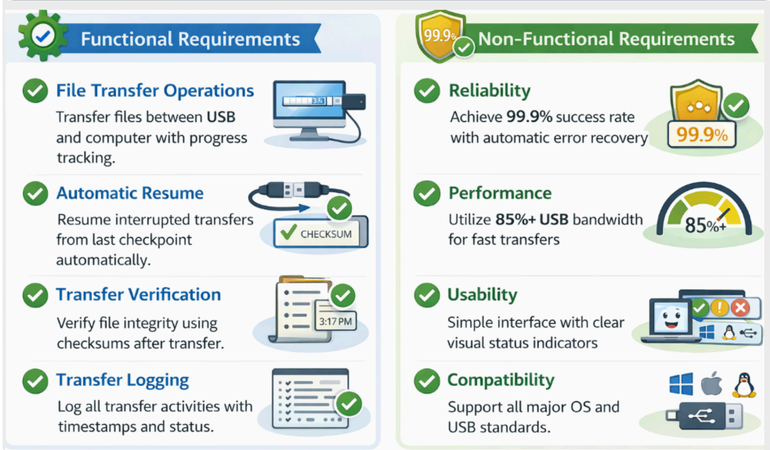
Table 1: Summary of Key Findings from Relevant Literature

| Author | Contribution | Limitation |
|---|---|---|
| USB-IF [1] | Defines USB standards ensuring reliable hardware-level data transfer. | No support for auto-resume at application level after interruption. |
| Tridgell et al. [2] | Introduced efficient partial file transfer using synchronization techniques. | Designed for network transfers, not direct USB devices. |
| Patterson et al. [3] | Proposed checkpoint-based fault tolerance for interrupted processes. | Complex and not optimized for lightweight USB file transfer systems. |
| Microsoft FS Team [4] | Improved file integrity using journaling file systems. | Cannot resume partially transferred files automatically. |

## Social Impact

- Reduces frustration and time loss for students, professionals, and general users during large file transfers.
- Improves productivity by ensuring dependable data handling in educational, office, and personal environments.
- Promotes digital reliability, encouraging wider adoption of efficient and user-friendly technology solutions.

## Software Requirements Specification




## SDLC

Table 2: Comparative Analysis of SDLC Models

| Priority | Criteria | Waterfall | V-Shape | Iterative | Spiral | Agile | Prototype | RUP |
|---|---|---|---|---|---|---|---|---|
| 3 | Well Known Requirement & Proper Documentation | Yes | No | No | No | No | No | No |
| 3 | Frequently Feedback Improved | No | No | Yes | Yes | Yes | No | Yes |
| 7 | Reliability and Efficiency | No | No | Yes | Yes | Yes | No | Yes |
| 4 | Risk Analysis | No | No | No | Yes | Yes | No | Yes |
| 4 | Re-use Components Capability | No | No | No | No | No | No | Yes |
| 4 | High User Involvement | No | No | No | Yes | Yes | No | No |
| 5 | Time Utilization | Yes | No | Yes | No | Yes | No | No |
| Total = 30 | | 8 | 0 | 15 | 18 | 23 | 0 | 18 |

## UI / UX

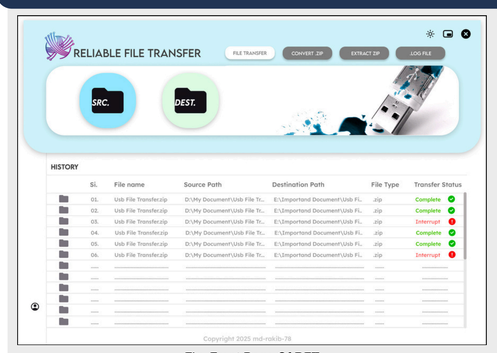



Fig: Front Page Of RFT

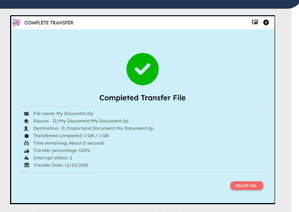



Fig: Completed Transfer History Page

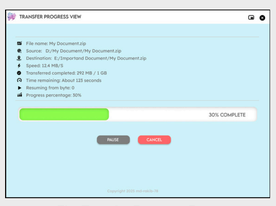


Fig: Transfer Progress Page

Fig: Transfer Cancel Warning Page

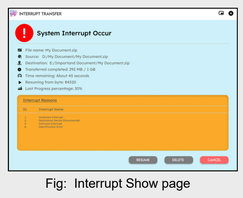



Fig: Interrupt Show page

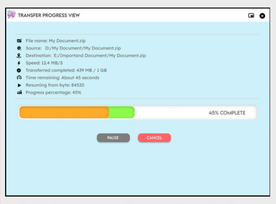


Fig: After Interrupt Resume Transfer Page

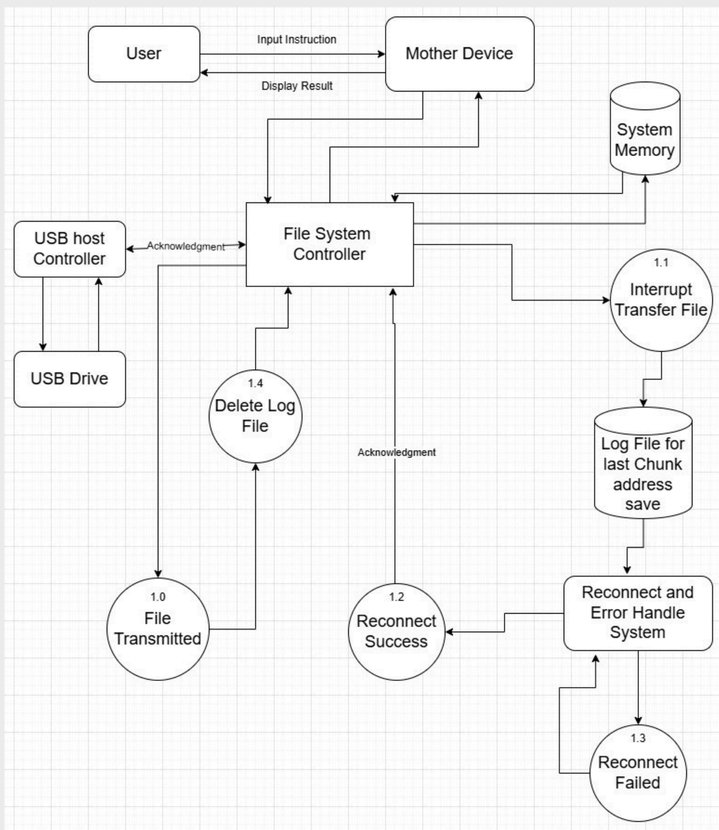
Fig: Log File History Page

Fig: Sign in Page

Fig: Login Page

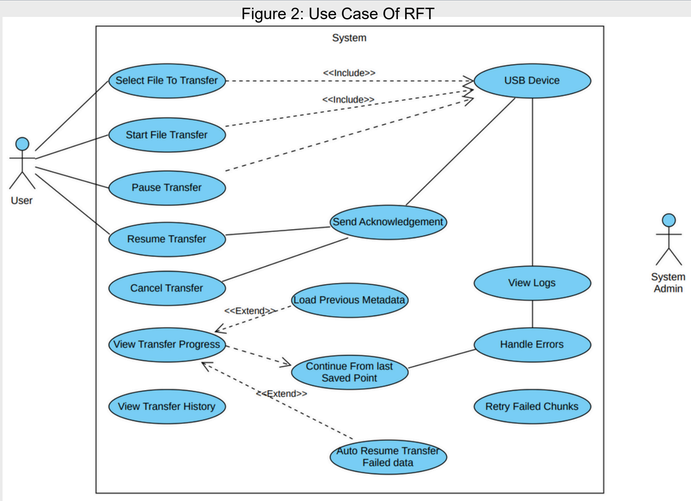
## Data Flow Diagram

Figure 1: DFD Level 2 Diagram Of RFT



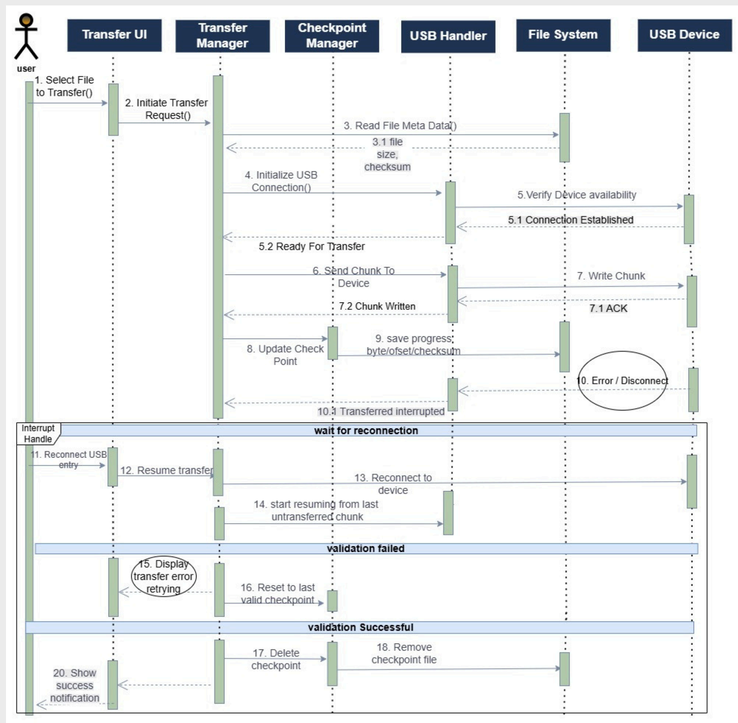

## Use Case Diagram

Figure 2: Use Case Of RFT



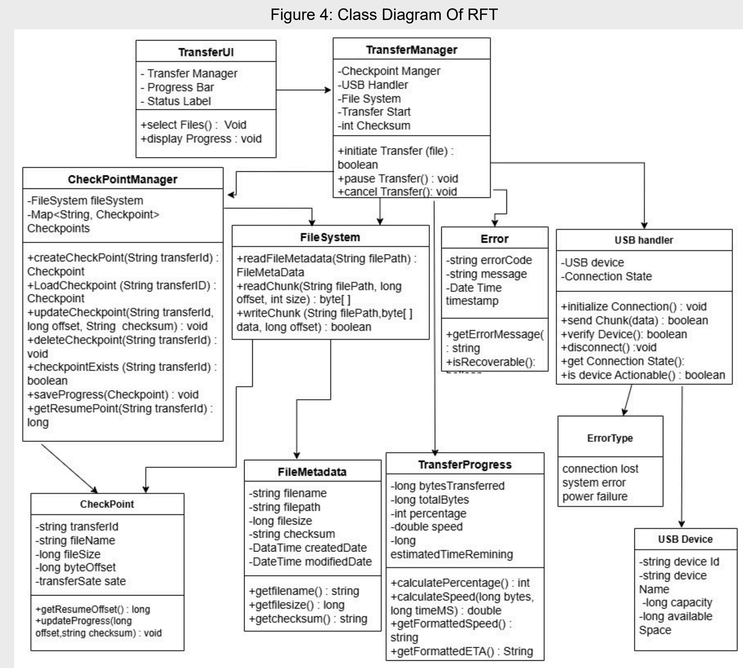

## Sequential & Communicational

Figure 3: Sequential & Communicational Of RFT




## Class Diagram

Figure 4: Class Diagram Of RFT




## Conclusion

The reliable USB file transfer system with auto resume support provides a robust solution for uninterrupted data transfer. By detecting interruptions and continuing from the last successful point, it saves time, reduces user frustration, and ensures efficiency, especially for large files. This system demonstrates a practical approach to fault-tolerant file management, enhancing overall user experience and reliability in everyday digital tasks.

## Reference


[1]. B Naresh Kumar Reddy “An Efficient Data Transmission by using Modern USB Flash Drive”

[2]. K. B. Tripathy, A. Sharma, P. Goyal, A. Sharma and V. Upmanu, "A Concept Of Data Transfer Via Bluetooth In Pen Drive", Journal of Global Research in Computer Science, 4(4), 2013. 22

[3]. Subhash Suman , "Data Transfer Between Two USB Disk Without Use Of Computer", International Journal Of Emerging Technology And Advanced Engineering, ISO9001:2008, Volume 3, Issue 5, May 2013.